Independent projectile directions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why horizontal and vertical projectile motion share time but evolve differently.

SUCCESS CRITERIA - I can...

1. Identify horizontal and vertical velocity components.
2. Use the same elapsed time for both axes.
3. State the negligible-air-resistance assumption.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “projectile” and “horizontal” to explain your model or result. Sentence starters are optional.

projectile: an object moving after launch under the stated forces

horizontal: along a level line, usually the x direction

vertical: up and down, usually the y direction

01 Warm-up: bring back an earlier idea

Outside a spherical source, field at r is 75 N/kg.
Find field at 5 r and the force on 0.7 kg there.

Word help: field = a description of an influence at each location

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $a_x = 0$; $a_y = -g$; v_x = constant.
Upward positive; g = 10 m/s². Ideal near-Earth motion with negligible air resistance.

Worked reasoning: A ball launches horizontally at 3 m/s and falls for 2 s. Horizontal travel is 3×2=6 m. With g=10 m/s² and no initial vertical velocity, its final vertical velocity is 20 m/s down.

Your turn: Draw separate horizontal and vertical velocity arrows at launch and after 2 s.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

One ball is dropped and another launched horizontally from the same height at the same instant.
Compare fall times and explain.

Sentence starter: Both … . They differ in … because … .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

A ball rolls horizontally off a platform at 8 m/s. It lands 1.5 s later.
Label the model and find horizontal displacement, vertical displacement and both final velocity components. Take upward as positive, use g = 10 m/s² and ignore air resistance.

Sentence starter: I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A student says a larger horizontal speed holds the ball up longer.
Explain which force or acceleration would be needed for that claim.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

For $x_i = 4 \pm 1$ m, $v = 6 \pm 0.5$ m/s and exact $t = 8$ s, find the central final position and conservative interval.
Name a specific check of your method.
Label the measured positions and the range each allows.

Word help: conservative bound = a limit chosen to include every allowed case

Sentence starter: I found … [unit]. This value means … in this situation.

Position and uncertainty (m)



Time (s)

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A projectile's velocity is (6,-8) m/s.
Find speed. What information is still needed to determine its position?
Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Sentence starter: I found … [unit]. This value means … in this situation.

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A projectile has initial velocity (6, 8) m/s; use upward positive and g=10 m/s².
Find velocity after 1 s and explain whether its horizontal motion depends on its vertical velocity.
Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Sentence starter: Under the assumption …, the evidence … supports … because … .



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3